

B9 Transformations

Name: _____

Class: _____

Date: _____

Time: **258 minutes**

Marks: **217 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

- (a) Use the trapezium rule with four ordinates (three strips) to find an approximate value for $\int_0^6 \sqrt{x^3 + 1} \, dx$, giving your answer to four significant figures. (4)
- (b) The curve with equation $y = \sqrt{x^3 + 1}$ is stretched parallel to the x -axis with scale factor $\frac{1}{2}$ to give the curve with equation $y = f(x)$. Write down an expression for $f(x)$. (2)
- (Total 6 marks)**

Q2.

- (a) Sketch the graph of $y = 3^x$, stating the coordinates of the point where the graph crosses the y -axis. (2)
- (b) Describe a single geometrical transformation that maps the graph of $y = 3^x$:
- (i) onto the graph of $y = 3^{2x}$; (2)
 - (ii) onto the graph of $y = 3^{x+1}$. (2)
- (c) (i) Using the substitution $Y = 3^x$, show that the equation $9^x - 3^{x+1} + 2 = 0$ can be written as $(Y - 1)(Y - 2) = 0$. (2)
- (ii) Hence show that the equation $9^x - 3^{x+1} + 2 = 0$ has a solution $x = 0$ and, by using logarithms, find the other solution, giving your answer to four decimal places. (4)
- (Total 12 marks)**

Q3.

- Describe a sequence of **two** geometrical transformations that maps the graph of $y = x^2$ onto the graph of $y = 4x^2 - 5$. (Total 4 marks)

Q4.

- (a) Express $x^2 - 3x + 4$ in the form $(x - p)^2 + q$, where p and q are rational numbers.

(2)

- (b) Hence write down the minimum value of the expression $x^2 - 3x + 4$.

(1)

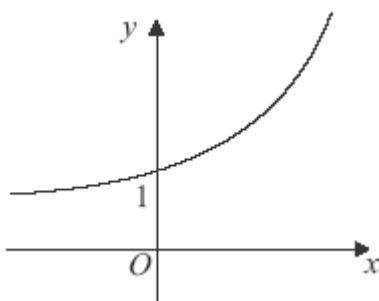
- (c) Describe the geometrical transformation that maps the graph of $y = x^2$ onto the graph of $y = x^2 - 3x + 4$.

(3)

(Total 6 marks)

Q5.

The diagram shows a sketch of the curve with equation $y = 6^x$.



- (a) (i) Use the trapezium rule with five ordinates (four strips) to find an approximate value for $\int_0^2 6^x dx$, giving your answer to three significant figures.

(4)

- (ii) Explain, with the aid of a diagram, whether your approximate value will be an overestimate or an underestimate of the true value of $\int_0^2 6^x dx$.

(2)

- (b) (i) Describe a single geometrical transformation that maps the graph of $y = 6^x$ onto the graph of $y = 6^{3x}$.

(2)

- (ii) The line $y = 84$ intersects the curve $y = 6^{3x}$ at the point A . By using logarithms, find the x -coordinate of A , giving your answer to three decimal places.

(4)

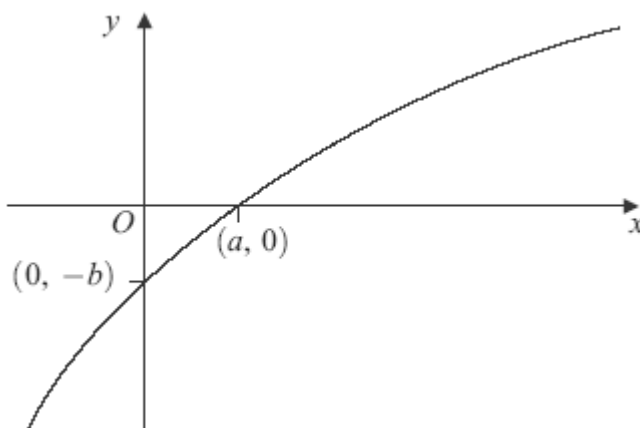
- (c) The graph of $y = 6^x$ is translated by $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$ to give the graph of the curve with equation $y = f(x)$. Write down an expression for $f(x)$.

(2)

(Total 14 marks)

Q6.

- (a) The diagram shows part of the curve with equation $y = f(x)$. The curve crosses the x -axis at the point $(a, 0)$ and the y -axis at the point $(0, -b)$.



On a separate diagram, sketch the curve with the equation $y = 2f(x)$. On the diagram, indicate, in terms of a or b , the coordinates of the points where the curve crosses the coordinate axes.

(2)

- (b) (i) Describe a sequence of geometrical transformations that maps the graph of $y = \ln x$ onto the graph of $y = 4 \ln(x + 1) - 2$.

(6)

- (ii) Find the exact values of the coordinates of the points where the graph of $y = 4 \ln(x + 1) - 2$ crosses the coordinate axes.

(4)

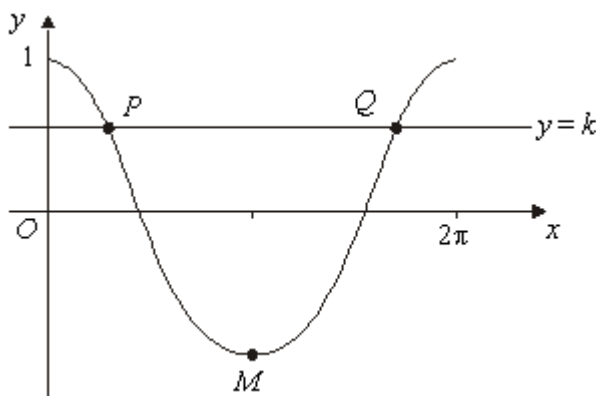
(Total 12 marks)

Q7.

- (a) Solve the equation $\cos x = 0.3$ in the interval $0 \leq x \leq 2\pi$, giving your answers in radians to three significant figures.

(3)

- (b) The diagram shows the graph of $y = \cos x$ for $0 \leq x \leq 2\pi$, and the line $y = k$.



The line $y = k$ intersects the curve $y = \cos x$, $0 \leq x \leq 2\pi$, at the points P and Q .
The point M is the minimum point of the curve.

- (i) Write down the coordinates of the point M .

(2)

- (ii) The x -coordinate of P is α .

Write down the x -coordinate of Q in terms of π and α .

(1)

- (c) Describe the geometrical transformation that maps the graph of $y = \cos x$ onto the graph of $y = \cos 2x$.

(2)

- (d) Solve the equation $\cos 2x = \cos \frac{4\pi}{5}$ in the interval $0 \leq x \leq 2\pi$, giving the values of x in terms of π .

(4)

(Total 12 marks)

Q8.

Describe a sequence of **two** geometrical transformations that maps the graph of $y = \sec x$ onto the graph of $y = 1 + \sec 3x$.

(Total 4 marks)

Q9.

- (a) Sketch the graph of $y = |2x|$.

(1)

- (b) On a separate diagram, sketch the graph of $y = 4 - |2x|$, indicating the coordinates of the points where the graph crosses the coordinate axes.

(3)

- (c) Solve $4 - |2x| = x$

(3)

- (d) Hence, or otherwise, solve the inequality $4 - |2x| > x$.

(2)

(Total 9 marks)

Q10.

- (a) (i) Express $x^2 + 10x + 19$ in the form $(x + p)^2 + q$, where p and q are integers.

(2)

- (ii) Write down the coordinates of the vertex (minimum point) of the curve with equation $y = x^2 + 10x + 19$. (2)
- (iii) Write down the equation of the line of symmetry of the curve $y = x^2 + 10x + 19$. (1)
- (iv) Describe geometrically the transformation that maps the graph of $y = x^2$ onto the graph of $y = x^2 + 10x + 19$. (3)
- (b) Determine the coordinates of the points of intersection of the line $y = x + 11$ and the curve $y = x^2 + 10x + 19$. (4)
- (Total 12 marks)

Q11.

- (a) Sketch the graph of $y = \tan x$ for $0^\circ \leq x \leq 360^\circ$. (3)
- (b) Write down the **two** solutions of the equation $\tan x = \tan 61^\circ$ in the interval $0^\circ \leq x \leq 360^\circ$. (2)
- (c) (i) Given that $\sin \theta + \cos \theta = 0$, show that $\tan \theta = -1$. (1)
- (ii) Hence solve the equation $\sin (x - 20^\circ) + \cos (x - 20^\circ) = 0$ in the interval $0^\circ \leq x \leq 360^\circ$. (4)
- (d) Describe the single geometrical transformation that maps the graph of $y = \tan x$ onto the graph of $y = \tan (x - 20^\circ)$. (2)
- (e) The curve $y = \tan x$ is stretched in the x -direction with scale factor $\frac{1}{4}$ to give the curve with equation $y = f(x)$. Write down an expression for $f(x)$. (1)
- (Total 13 marks)

Q12.

- (a) Describe the geometrical transformation that maps the curve with equation $y = \sin x$ onto the curve with equation:
- (i) $y = 2 \sin x$;

(2)

(ii) $y = -\sin x$;

(2)

(iii) $y = \sin(x - 30^\circ)$.

(2)

- (b) Solve the equation $\sin(\theta - 30^\circ) = 0.7$, giving your answers to the nearest 0.1° in the interval $0^\circ \leq \theta \leq 360^\circ$.

(3)

- (c) Prove that $(\cos x + \sin x)^2 + (\cos x - \sin x)^2 = 2$.

(4)

(Total 13 marks)

Q13.

- (a) Express $x^2 + 8x + 19$ in the form $(x + p)^2 + q$, where p and q are integers.

(2)

- (b) Hence, or otherwise, show that the equation $x^2 + 8x + 19 = 0$ has no real solutions.

(2)

- (c) Sketch the graph of $y = x^2 + 8x + 19$, stating the coordinates of the minimum point and the point where the graph crosses the y -axis.

(3)

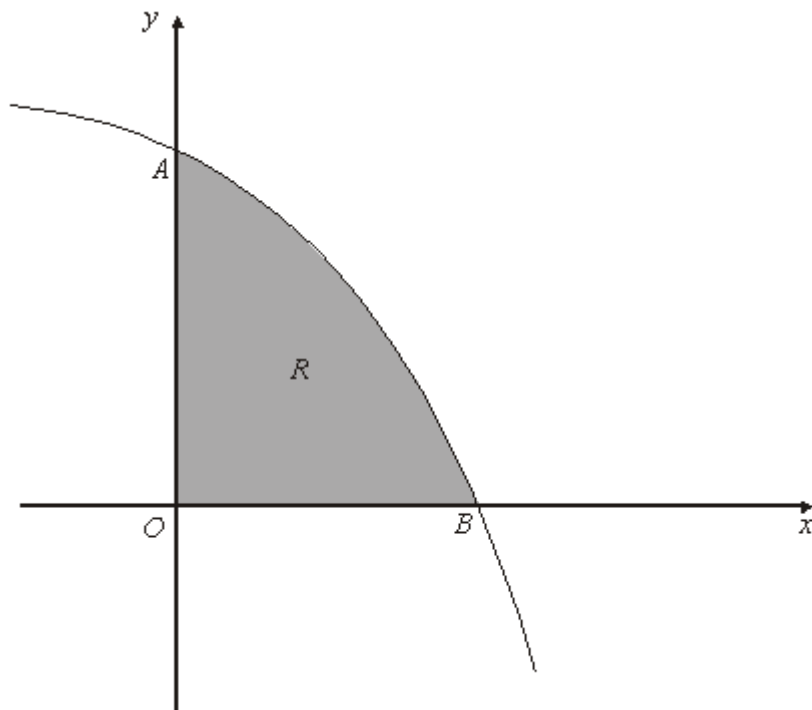
- (d) Describe geometrically the transformation that maps the graph of $y = x^2$ onto the graph of $y = x^2 + 8x + 19$.

(3)

(Total 10 marks)

Q14.

The diagram shows a sketch of the curve with equation $y = 27 - 3^x$.



The curve $y = 27 - 3^x$ intersects the y -axis at the point A and the x -axis at the point B .

- (a) (i) Find the y -coordinate of point A . (2)
- (ii) Verify that the x -coordinate of point B is 3. (1)
- (b) The region, R , bounded by the curve $y = 27 - 3^x$ and the coordinate axes is shaded. Use the trapezium rule with four ordinates (three strips) to find an approximate value for the area of R . (4)
- (c) (i) Use logarithms to solve the equation $3^x = 13$, giving your answer to four decimal places. (3)
- (ii) The line $y = k$ intersects the curve $y = 27 - 3^x$ at the point where $3^x = 13$.
Find the value of k . (1)
- (d) (i) Describe the single geometrical transformation by which the curve with equation $y = -3^x$ can be obtained **from** the curve $y = 27 - 3^x$. (2)
- (ii) Sketch the curve $y = -3^x$. (2)

(Total 15 marks)

Q15.

- (a) Describe the single geometrical transformation by which the curve with equation $y = \tan \frac{1}{2}x$ can be obtained from the curve $y = \tan x$.

(2)

- (b) Solve the equation $\tan \frac{1}{2}x = 3$ in the interval $0 < x < 4\pi$, giving your answers in radians to three significant figures.

(4)

- (c) Solve the equation

$$\cos \theta (\sin \theta - 3 \cos \theta) = 0$$

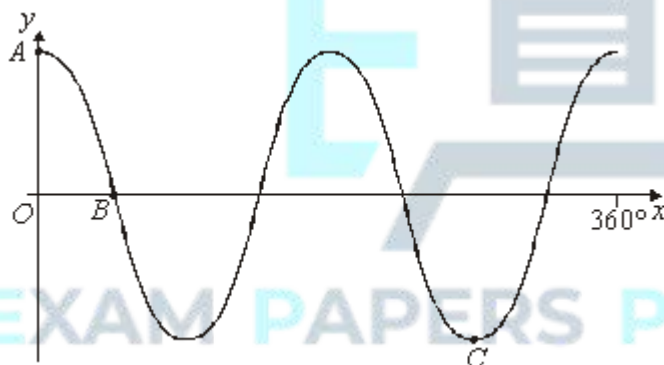
in the interval $0 < \theta < 2\pi$, giving your answers in radians to three significant figures.

(5)

(Total 11 marks)

Q16.

The diagram shows the graph of $y = \cos 2x$ for $0^\circ \leq x \leq 360^\circ$.



- (a) Write down the coordinates of the points A , B and C marked on the diagram.
- (b) Describe the single geometrical transformation by which the curve with equation $y = \cos 2x$ can be obtained from the curve with equation $y = \cos x$.
- (c) Solve the equation

(4)

(2)

$$\cos 2x = 0.37$$

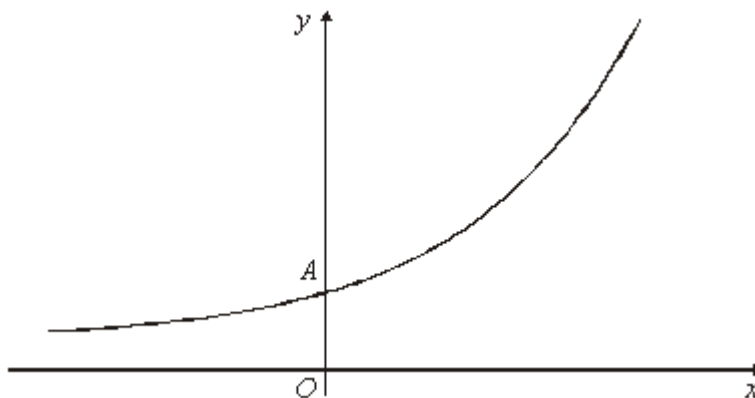
giving all solutions to the nearest 0.1° in the interval $0^\circ \leq x \leq 360^\circ$.
(No credit will be given for simply reading values from a graph.)

(5)

(Total 11 marks)

Q17.

The diagram shows a sketch of the curve with equation $y = 3^x + 1$.



The curve intersects the y -axis at the point A .

- (a) Write down the y -coordinate of point A . (1)
- (b) (i) Use the trapezium rule with five ordinates (four strips) to find an approximation for $\int_0^1 (3^x + 1) dx$, giving your answer to three significant figures. (4)
- (ii) By considering the graph of $y = 3^x + 1$, explain with the aid of a diagram whether your approximation will be an overestimate or an underestimate of the true value of $\int_0^1 (3^x + 1) dx$. (2)
- (c) The line $y = 5$ intersects the curve $y = 3^x + 1$ at the point P . By solving a suitable equation, find the x -coordinate of the point P . Give your answer to four decimal places. (4)
- (d) The curve $y = 3^x + 1$ is reflected in the y -axis to give the curve with equation $y = f(x)$. Write down an expression for $f(x)$. (1)

(Total 12 marks)

Q18.

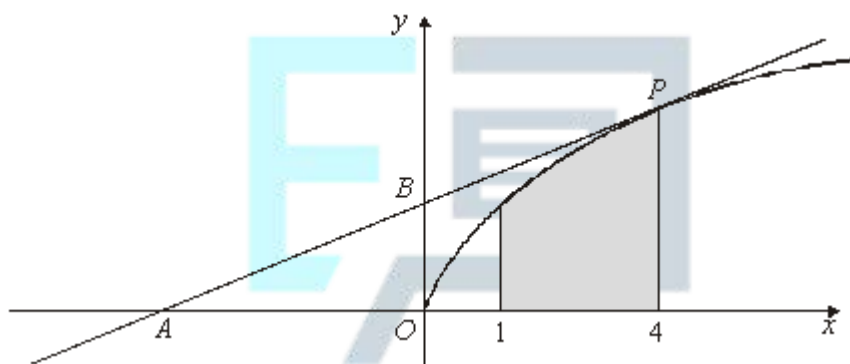
- (a) Express $x^2 - 6x + 16$ in the form $(x - p)^2 + q$. (2)
- (b) A curve has equation $y = x^2 - 6x + 16$. Using your answer from part (a), or

otherwise:

- (i) find the coordinates of the vertex (minimum point) of the curve; (2)
 - (ii) sketch the curve, indicating the value where the curve crosses the y -axis; (2)
 - (iii) state the equation of the line of symmetry of the curve. (1)
- (c) Describe geometrically the transformation that maps the graph of $y = x^2$ onto the graph of $y = x^2 - 6x + 16$. (3)
- (Total 10 marks)

Q19.

The diagram shows a curve C with equation $y = \sqrt{x}$. The point O is the origin $(0, 0)$.



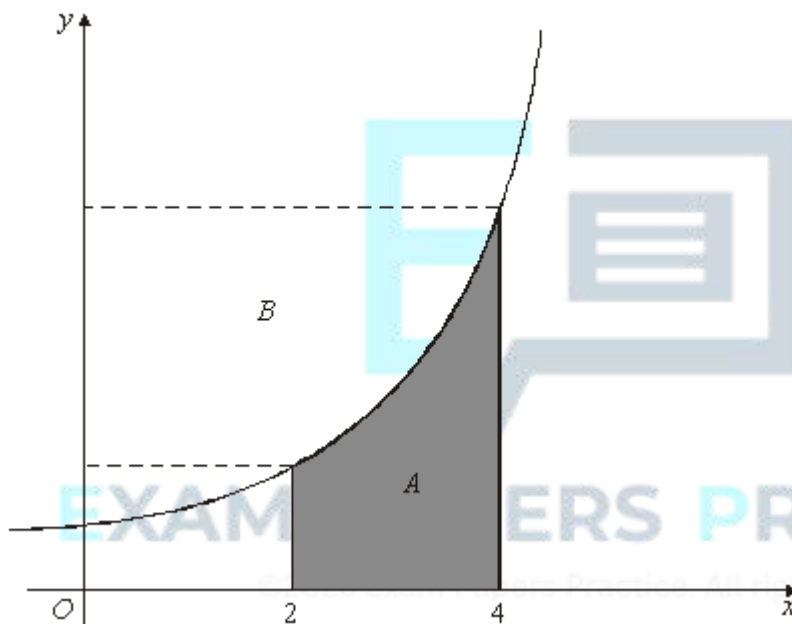
The region bounded by the curve C , the x -axis and the vertical lines $x = 1$ and $x = 4$ is shown shaded in the diagram.

- (a) (i) Write $\sqrt{x}$ in the form x^p , where p is a constant. (1)
 - (ii) Find $\int \sqrt{x} \, dx$. (2)
 - (iii) Hence find the area of the shaded region. (3)
- (b) The point on C for which $x = 4$ is P . The tangent to C at the point P intersects the x -axis and the y -axis at the points A and B respectively.
- (i) Find an equation for the tangent to the curve C at the point P . (4)

- (ii) Find the area of the triangle AOB . (3)
- (c) Describe the single geometrical transformation by which the curve with equation $y = \sqrt{x-1}$ can be obtained from the curve C . (2)
- (d) Use the trapezium rule with four ordinates (three strips) to find an approximation for $\int_1^4 \sqrt{x-1} \, dx$, giving your answer to three significant figures. (4)
- (Total 19 marks)

Q20.

The diagram shows part of the graph of $y = e^{2x} + 3$.



- (a) Describe a sequence of two geometrical transformations that maps the graph of $y = e^x$ onto the graph of $y = e^{2x} + 3$. (4)
- (b) Find the exact value of the area of the shaded region A. (4)
- (c) The region B is indicated on the diagram.
- Find the area of the region B, giving your answer in the form $pe^8 + qe^4$, where p and q are numbers to be determined. (4)

(Total 12 marks)